

Design Thinking in Business Processes and Startups

Design Thinking applied in Business & Strategic Innovation:-

Design Thinking is a human-centered, iterative approach to problem-solving that has gained widespread adoption in business and strategic innovation. It enables organizations to address complex challenges, drive creativity, and develop customer-focused solutions. Unlike traditional business methodologies that rely heavily on data-driven decision-making alone, Design Thinking incorporates empathy, experimentation, and collaboration to create innovative and effective strategies. Design Thinking helps in understanding customer needs and solving complex problems in an user-centric manner. Integrating Design thinking in Business enhances customer experiences, drives innovation, fosters collaboration, increased organizational agility, and sustained competitive advantage This approach is particularly valuable in today's fast-changing business environment, where adaptability and customer-centricity are key to maintaining a competitive edge.

Functional Steps involved in Business Innovation

- **Start small:** Begin with a pilot project applying Design Thinking principles to solve a specific problem
- **Encourage Cross-functional teams:** Include diverse members from different departments to bring varied perspectives to the table
- **Empower Employees:** Train employees on Design Thinking methodologies and give them the autonomy to explore creative solutions
- **Leverage Customer feedback:** Regularly gather insights from customers to refine products and services continually
- **Measure Impact:** Track the success of Design Thinking initiatives through metrics like customer satisfaction, time-to-market and revenue growth

Principles of Design Thinking in Business & Strategic Innovation

1. Customer-Centric Innovation and Enhanced User experience

One of the core principles of Design Thinking understands user needs and experiences. Businesses use this approach to:

- Conduct user research and identify customer pain points.
- Develop solutions that are not only functional but also emotionally engaging.
- Foster customer loyalty and satisfaction by offering products and services that truly meet their needs.

For example, companies like **Nike and Apple** incorporate Design Thinking to create products that offer seamless user experiences, blending technology and design to meet customer expectations effectively.

2. Agility and adaptable Business Strategies

Design Thinking enables organizations to move away from rigid, long-term strategic planning and adopt a more flexible and adaptive approach. Businesses benefit from:

- Rapid iteration based on real-time feedback and market insights.

- Quick adaptation to emerging trends, technological advancements, and customer expectations.
- A fail-fast mentality that encourages testing new ideas without the fear of making mistakes.

For example, **Amazon** continuously experiments with new business models, adjusting its strategies based on customer feedback and market conditions.

3. Increased innovation and Enhanced Competitive Advantage

In highly competitive markets, businesses need to differentiate themselves by providing unique value propositions and innovative designs or prototypes. Design Thinking allows organizations to:

- Focus on problem-solving from the user's perspective, leading to highly tailored solutions.
- Leverage creativity to innovate beyond traditional offerings.
- Introduce meaningful differentiation that is difficult for competitors to replicate.

Companies like **Airbnb** used Design Thinking to transform the hospitality industry by creating a personalized and community-driven alternative to traditional hotels.

4. Driving Digital Transformation

As businesses undergo digital transformation, Design Thinking ensures that technology solutions are user-friendly and aligned with customer needs. Key benefits include:

- Improving the user experience (UX) of digital platforms, applications, and services.
- Reducing resistance to digital adoption by ensuring intuitive and accessible interfaces.
- Helping companies like **IBM and Google** create customer-centric digital products that align with real-world use cases.

5. Solving Complex Business Problems and focusing on Risk Mitigation

Many businesses face challenges related to inefficiencies, outdated processes, and product-market misalignment. Design Thinking helps by:

- Identifying root causes instead of addressing surface-level symptoms.
- Encouraging cross-functional teams to brainstorm and prototype potential solutions.
- Implementing changes that are iterative and data-informed.

For example, **GE Healthcare** used Design Thinking to improve the experience of pediatric MRI scans, transforming an intimidating medical procedure into a more engaging and less stressful experience for children.

6. Encouraging Cross-Functional Collaboration and Teamwork

Innovation thrives when diverse teams work together to tackle challenges. Design Thinking fosters collaboration by:

- Developing teamwork among the technical and functional teams involved
- Breaking down organizational bottlenecks and encouraging knowledge sharing.
- Enabling teams from different disciplines—such as marketing, engineering, finance, and design—to co-create solutions.
- Enhancing internal engagement and alignment towards shared business goals.

Companies like **PepsiCo** have successfully used Design Thinking to integrate different departments in the innovation process, leading to more holistic and impactful business solutions.

7. Rapid Prototyping & Experimentation

Instead of investing large amounts of resources into untested ideas, businesses use Design Thinking to develop prototypes and run small-scale experiments before full-scale implementation. This approach helps to:

- Reduce financial and operational risks.
- Gather valuable user feedback to refine and improve solutions.
- Ensure that final products or services are well-aligned with market needs.

For example, **Tesla** continually refines its electric vehicles based on real-world testing and customer input, allowing for rapid innovation and product enhancement.

8. Accelerated Innovation Cycle: The iterative nature of Design Thinking reduces time-to-market by identifying issues early in the development cycle, allowing quick adjustments

Real-World Case Studies

- **Apple:** Uses Design Thinking to create user-friendly, aesthetically pleasing, and highly functional products that dominate the market.
- **IBM:** Integrated Design Thinking into its corporate culture, leading to increased innovation and improved product development cycles.
- **Airbnb:** Overcame early struggles by reimagining the customer journey and improving the overall user experience through a Design Thinking approach.
- **GE Healthcare:** Used Design Thinking to redesign pediatric MRI machines, making them more child-friendly and reducing anxiety for young patients.

Principles of Design Thinking That Redefine Business:-

Design Thinking is built upon several core principles that guide businesses towards innovative and effective problem-solving. These principles help organizations develop user-centered solutions, foster a culture of innovation, and remain competitive in rapidly evolving markets.

1. Empathy-Driven Problem Solving

Empathy is the foundation of Design Thinking, requiring businesses to deeply understand their customers' needs, pain points, and behaviors. This is achieved through:

- Conducting qualitative and quantitative research.
- Engaging directly with users through interviews, surveys, and observations.
- Developing personas and journey maps to visualize user experiences.

By prioritizing empathy, companies can design solutions that truly resonate with their target audiences.

2. Iterative and Experimentation-Based Approach

Unlike traditional linear problem-solving methods, Design Thinking is an iterative process that encourages continuous refinement of ideas. Key aspects include:

- Rapid prototyping to test concepts before full-scale implementation.
- Gathering feedback from users and making incremental improvements.

- Encouraging a mindset of learning from failure rather than fearing it.

This approach allows businesses to reduce risks and increase the likelihood of success.

3. Human-Centered Design

Every solution developed through Design Thinking must be centered around the end-user. Businesses focus on:

- Prioritizing user needs over internal operational efficiency.
- Designing intuitive and accessible experiences.
- Creating solutions that provide both functional and emotional value.

Companies like **Apple and Airbnb** have mastered human-centered design by making their products and services deeply engaging and easy to use.

4. Cross-Disciplinary Collaboration

Innovation flourishes when diverse teams work together. Design Thinking promotes collaboration across departments such as:

- Marketing, technology, finance, and design teams.
- Encouraging an open exchange of ideas and perspectives.
- Breaking down organizational silos to foster teamwork and co-creation.

IBM and PepsiCo have successfully integrated cross-functional collaboration into their innovation processes.

5. Action-Oriented Mindset

Instead of overanalyzing problems, Design Thinking encourages action and rapid execution through:

- Building prototypes and minimum viable products (MVPs).
- Testing solutions with real users early in the development cycle.
- Making data-driven decisions based on real-world feedback.

This principle ensures that businesses move forward efficiently and pivot when necessary.

Business Challenges

Organizations face various challenges that test their capability, resilience and innovation. There are different challenges faced by the Industry. The typical challenges anticipated from the organizations are summarized in the Fig. below.

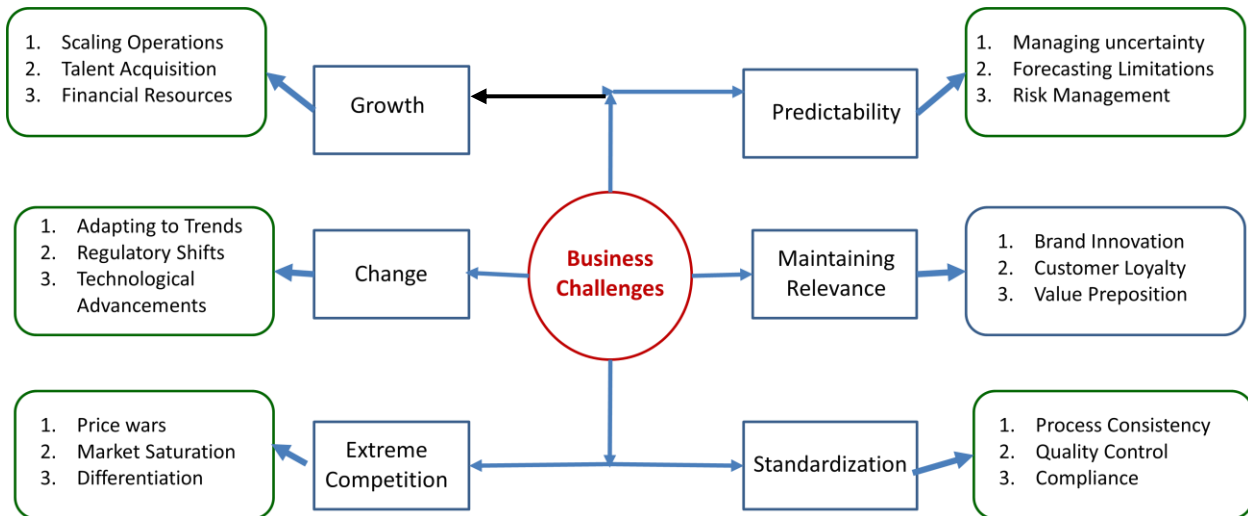


Fig. Business Challenges

Design Thinking in Predicting Business Challenges

S No	Approach	Description
1	Empathy	Empathy with users and stakeholders allows businesses anticipate challenges before they emerge by identifying unmet needs
2	Proactive Problem identification	Thorough examination through the Customer Journey maps uncovers important and sensitive problems. This insight can help to take proactive steps and reduce gaps, predicting customer behavior changes etc.
3	Flexibility and adaptability	Design thinking approach helps in encouraging adaptability and makes continuous improvements. This is essential to responding and predicting challenges and make improvements
4	Culture of Experimentation	By testing ideas and solutions at an early stage, businesses can identify the potential roadblocks and adapt their strategies accordingly
5	Better Decision making	The into problem-solving process cross-functional collaboration adopted helps in bringing diverse perspectives which leads to improved decision making
6	Alignment with Customer Trends and Market needs	Through regular customer feedback and research, the real time insights obtained are aligned with the changing trends and market shifts for better reach of customers

Business Models

A business model is a strategic plan of how a company will make money. The model describes the way a business will take its product, offer it to the market, and drive sales.

Example: Consider the vast portfolio of Microsoft. Over the past several decades, the company has expanded its product line across digital services, software, gaming, and more. Various business models, all within Microsoft, include but are not limited to:

- **Productivity and business processes:** Microsoft offers subscriptions to Office products and LinkedIn. These subscriptions may be based on product usage (i.e. the amount of data being uploaded to SharePoint).
- **Intelligent cloud:** Microsoft offers server products and cloud services for a subscription.
- **Personal computing:** Microsoft sells the Windows operating system as well as physically manufactured products such as Surface, PC components, and Xbox hardware. Residual Xbox sales include content, services, subscriptions, royalties, and advertising revenue.

Components of Business Models

S No	Component	Description
1	Value Proposition	It defines why a customer should choose one product or service or others. A compelling value proposition clearly articulates benefits, unique features and differentiators
2	Customer Segments	Identifying specific groups of customers who will benefit from the product or service is critical. Understanding the demographics, preferences, and behaviors of these segments allows for targeted marketing and tailored offerings
3	Revenue streams	It outlines how the business generates income such as direct sales, subscriptions, licensing, advertising etc
4	Cost structure	Understanding the costs associated with running the business including fixed and variable costs, is crucial. It helps identify key expenses and potential areas for cost reduction.
5	Key Resources	Resources that will deliver at the end such as human resources, technology, intellectual property and physical assets are covered.
6	Key activities	This refers to the essential actions and processes required to deliver the value proposition, reach customers, and maintain customer relationships
7	Channels	The avenues both offline and online, through which the company delivers its value proposition to customers. Channels may include physical stores, e-commerce platforms, social media and distribution partnerships
8	Customer Relationships	Understanding how to engage with customers and build loyalty is vital. This can range from personalized service and community engagement to automated support systems
9	Key Partnerships	Collaborations with suppliers, distributors, and strategic partners can enhance the business model. Partnerships can provide access to resources, technology or markets that may be challenging to achieve independently

Testing of Business Models

S No	Component	Description
1	Market Research	Conducting surveys, focus groups, and interview with potential customers help gather insights into their needs, preferences and willingness to pay. This data helps to make adjustments to value proposition and overall model
2	Pilot Programs and MVPs	Creating Minimum Value Products (MVPs) allows startups to test their concepts in the market with minimal investment. MVPs are simplified versions of the product that include just enough features to meet initial customer needs
3	Feedback Loops	Regularly soliciting feedback from users enables continuous improvement. This can be done through surveys, user testing sessions, and analytics to assess how customers interact with the product
4	Metrics and KPIs	Establishing Key Performance Indicators helps measure the success of business model. Metrics can include customer acquisition cost, lifetime value, churn rate, and conversion rates providing actionable insights
5	Iteration	Based on feedback and performance data, businesses should be prepared to iterate on their model for refining the value proposition, adjusting the price etc

Defining and Testing Business Models and Business Cases:-

Design Thinking plays a crucial role in defining and testing business models and business cases. It helps companies develop sustainable, scalable, and customer-centered business strategies by integrating experimentation and user feedback. A business model outlines how a company creates, delivers, and captures value. Design Thinking helps businesses identify customer needs and tailor value propositions accordingly. A business case justifies the viability of a project or investment.

Design Thinking allows businesses to prototype and validate concepts before committing significant resources. Startups and enterprises alike use Design Thinking to develop Minimum Viable Products (MVPs). MVPs help validate assumptions and reduce risks before full-scale launch. Business models must evolve based on market conditions and customer insights. Design Thinking fosters a mindset of agility and responsiveness.

Examples of few Business models:

- **Uber's business model** was shaped by iterative testing and feedback, ensuring that ride-sharing met user needs effectively.
- **Airbnb tested its business case** by offering short-term stays in its founders' apartment, refining the model based on early user feedback.
- **Dropbox's MVP** used a simple explainer video to test market demand before building a complex platform.
- **Netflix's** evolution from DVD rentals to a streaming service showcases the importance of iterative business model testing.

Business Model – Example: Assume a smart watch with integrated AI features for health monitoring is designed and a business model is to be developed. The following steps are to be followed:

- Step-1: Outline the ideas through Business Canvas Map
- Step-2: Identify the key partners including manufacturers, software developers and health advisors
- Step-3: Identify the possible revenue streams and channels, both offline and online
- Step-4: Decide to ensure the users require it and buy it
- Step-5: Conduct Empathy based interviews for the insights of customers and health awareness
- Step-6: Analyze the data – how many are willing to buy or subscribe, ceiling price etc.
- Step-7: Adjust the Business model to suit the data
- Step-8: Develop strategies for the investors, about the highlights of the product
- Step-9: Test the prototype and take the initial feedback
- Step-10: Analyze the feedback
- Step-11: Highlight the success stories of prototype advantages through physical models and social media
- Step-12: Attract the investors with all the information and make them convinced
- Step-13: Launch the product successfully
- Step-14: Collect the continuous feedback and upgrade the versions of the product for customer acceptance

Startups

It is a company in its early stages of operation, typically characterized by high growth potential and a focus on innovation or disruption, often requiring significant investment to get off the ground. It is a newly created company that relies on information and communication technologies (ICT) to market its products or services. It features a scalable business model in which its priority is to grow quickly and smoothly. It is a newly formed business with particular momentum behind it based on perceived demand for its product or service. During the beginning, startups face high uncertainty and have high rates of failure, but a few of them go on to become successful and influential – called as Unicorns.

What is the Startup India Initiative?

Launched on 16th January, 2016, the Startup India Initiative has rolled out several programs with the objective of supporting entrepreneurs, building a robust startup ecosystem and transforming India into a country of job creators instead of job seekers. These programs are managed by a dedicated Startup India Team, which reports to The Department for Promotion of Industry and Internal Trade (DPIIT). Under the Startup India initiative, eligible companies can get recognized as Startups by DPIIT, in order to access a host of tax benefits, easier compliance, IPR fast-tracking and more.

Typical Startups in India

 CRED	 Razorpay	 1mg
 Boat	 Delhivery	 Ola Cabs
 Swiggy	 upGrad	 Groww
 Meesho	 Nykaa	 Paytm
 PharmEasy	 Udaan	 Unacademy
 Urban Company	 Digit Insurance	 Livspace
 Skyroot Aerospace	 Supersourcing	 Zepto

The current value of some of the Startups in India is given below:

- **BYJU'S**: Edtech giant with a valuation of \$16 billion.
- **Flipkart**: E-commerce platform owned by Walmart, with a valuation of \$37.6 bn
- **PhonePe**: Digital payments platform, with a valuation of \$12 billion.
- **Swiggy**: Food delivery platform, with a valuation of \$8 billion.
- **Dream11**: Fantasy sports platform, with a valuation of \$8 billion.
- **Razorpay**: Fintech startup, with a valuation of \$7.5 billion.
- **Ola Electric**: Electric mobility company, with a valuation of \$4 billion.
- **Paytm**: Digital payments and financial services platform, valuation of \$16 billion.
- **Udaan**: B2B e-commerce platform, with a valuation of \$3.1 billion.
- **Nykaa**: E-commerce platform for beauty and fashion, with a valuation of \$8 billion.

Design Thinking for Startups:-

Startups can leverage Design Thinking to navigate uncertainty, identify market opportunities, and develop innovative solutions quickly. Key benefits include:

1. Validating Ideas Rapidly

- Startups can use Design Thinking to test assumptions and validate ideas before full-scale development.
- Example: Dropbox used a simple video prototype to gauge interest before developing its product.

2. Customer-Centric Product Development

- Helps startups build solutions that align with actual user needs.
- Example: Airbnb refined its platform through user feedback and iteration.

3. Pivoting with Agility

- Design Thinking fosters adaptability, enabling startups to pivot based on market demands.
- Example: Instagram pivoted from a check-in app to a photo-sharing platform.

Success stories of Selected Startups:

Company name: OYO

Started: 2013

Founder: Ritesh Agarwal

Industry: Hospitality

Net Valuation: \$16 billion

- Ritesh Agarwal, the founder of unicorn Startup Oyo, has made Oyo one of the largest hotel chains in the world.
- They currently operate in more than 800 urban areas across countries like India, China, Brazil, Japan, United States and 80+ more.
- Airbnb was the greatest motivation for Ritesh Agarwal, who began his journey as a traveler and when he found a huge problem in the hotels.
- He took it as an opportunity to become an entrepreneur by solving that problem.

Company name: Paytm

Started: 2010

Founder: Vijay Shekhar Sharma

Industry: Fintech

Net Valuation: \$15 billion

- Paytm began as a digital wallet in its early days and not yet had changed into what it is today.
- The fintech administration has had the option to arise as the top fintech organization even in the wake of confronting savage early rivalry from brands like Freecharge.
- Paytm's user base developed from 125 million to 185 million three months post demonetization.
- Vijay Shekhar Sharma, who was battling to make a decent living with Rs 10 in his pocket, tasted triumph the most difficult way possible, and today, he remains as the author of the billion-dollar local unicorn.

Company name: Flipkart

Started: 2007

Originators: Sachin Bansal and Binny Bansal

Industry: Ecommerce

Assessed Valuation: \$24 billion

- The Walmart obtained online business organization flipkart remains as the biggest web based business stage in India.
- Today, the web based business stage has extended its business skylines by foraying into the video web based industry with Flipkart Video Originals.
- Sachin Bansal and Binny Bansal began as an online book shop that made around 20 effective shipments in its first year of working.
- It was not late before the stage began catching individuals' eye that made it the top online retail market ever.

Company name: OLA

Started: 2010

Organizers: Bhavish Aggarwal, Ankit Bhati

Industry: Mobility

Assessed Valuation: \$6.2 billion

- Ola Cabs, begun by IIT-B graduates Bhavish and Ankit, has arisen to be the most conspicuous Indian portability specialist organization in the country that offers its administrations across 50 urban areas in India and is equally spread across nations, including the UK, Australia, and New Zealand.
- The possibility of Ola, a taxi total startup, came from Bhavish's horrendous experience. Simultaneously, he was venturing out from Bengaluru to Bandipur, where the driver of his taxi abandoned him in no place over a bombed arrangement bargain. This is a good startup Stories.

Benefits of Design Thinking to Startups

- **Understanding and Addressing Customer needs**
 - User Research, Empathy mapping and Customer Journey Maps helps in identifying the problems
- **Validating ideas quickly and Cost-effectively**
 - Ideation sessions, Rapid prototyping and Feedback loops helps in validating effective ideas
- **Achieving Product Market Fit**
 - Problem framing, User-Centric solutions, and Iterative testing are important in this category
- **Creating a Differentiated Brand experience**
 - Brand storytelling, Designing for Delight and Touch point optimization are essential in this category
- **Encouraging a Lean, Agile Culture**
 - Fail first and Learn fast, Cross-functional Collaboration, Responsive Development are important in this category
- **Enhancing Problem solving Capabilities**
 - Reframing problems, Structured Brainstorming, Data-driven solutions form the core in this category
- **Scaling, Loyalty and Sustainability**
 - User feedback integration, Flexible scalability, Sustainable practices, Increasing Retention, Value creation are important aspects

Developing Prototypes

Objectives

- To create a representation of the product or service that can be tested and iterated upon
- To select the type of prototype once the goals of the product or service are outlined
- To create the prototype using tools such as Sketch, Figma, Adobe XD or any physical materials
- To focus on the key functionalities and design elements
- To develop a team with diverse skills and have a collaborative work atmosphere

- To gather enough financial resources for a successful output
- To create different versions of prototypes based on feedback and refine as per user needs

Major Steps involved in developing prototypes:

- Define Objectives
- Choose type of Prototypes – Low fidelity, Medium fidelity or high Fidelity
- Create the prototype
- Gather resources
- Iterate quickly

Testing Prototypes

- **Define testing goals**
 - Clearly define what you want to test and what you want to learn from this step
- **Select testing methods**
 - Usability testing: Involves observing users as they interact with the prototype. This can be done in-person or remotely using techniques such as think-aloud protocols where users articulate their thoughts as they navigate
 - A/B testing: Compares two versions of a prototype to see which one performs better on specific metrics such as user engagements or conversion rates
 - Surveys and Questionnaire: Collect quantitative and qualitative feedback from users
- **Recruit test participants**
 - Identify and recruit individuals who represent the target audience. Their feedback will provide insights into the products usability and appeal
- **Conduct testing sessions**
 - Facilitate testing sessions in a controlled environment where users can freely interact with the prototype. Encourage open dialogue and ensure participants feel comfortable sharing their thoughts and experiences
- **Analyze results**
 - Collect and analyze the data from testing sessions, looking for patterns in user behavior, preferences and pain points. Use tools for qualitative and quantitative analysis selectively
- **Iterate based on feedback**
 - Use the insights gained from testing to refine and improve the prototype. Address any identified issues and make adjustments to features or design elements
- **Repeat testing**
 - Prototyping is an iterative cycle. Continue testing and refining until the prototype effectively meets user needs and aligns with business objectives

Benefits of Prototyping and Testing

- **Early detection of the problems**
 - Identification of design flaws or usability issues early reduces costly changes later on
- **Enhanced User Engagement**

- Involving users in the testing phase fosters a sense of ownership and can lead to valuable insights that drive design improvements
- **Refined Product Design**
 - Iterative testing and feedback enable teams to refine the product continually enhancing its appeal and usability
- **Risk reduction**
 - Validating concepts before full-scale development can minimize the risk of failure
- **Improved communication**
 - Prototypes serve as visual communication tools that help stakeholders understand the product vision, fostering alignment among team members

Technology Readiness Levels (TRLs) and Manufacturing Readiness Levels (MRLs)

The TRL is a system for assessing the maturity of a technology, ranging from basic research (TRL1) to fully operational deployment (TRL9). The TRL is used to track the progress of technology development and ensure consistent evaluation across different fields. The MRL is a measure to assess the maturity of manufacturing readiness, similar to how technology readiness levels are used for technology readiness. The MRLs can be used in general industry assessments, or for more specific application in assessing capabilities of possible suppliers.

These levels are important for monitoring when the startups are proposed to be launched into the market and for continuous assessment of their status at each stage. The details are illustrated in the Figs. below in the same order.



Fig. Technology Readiness Levels

Material Solutions Analysis	1	Basic manufacturing implications identified.
	2	Manufacturing concepts identified
	3	Manufacturing proof-of-concept developed
	4	Capability to produce the technology in a laboratory environment
Technology Development	5	Capability to produce prototype components in a production relevant environment
	6	Capability to produce a prototype system or subsystem in a production relevant environment
Engineering and Manufacturing Development	7	Capability to produce systems, subsystems or components in a production representative environment
	8	Pilot line capability demonstrated. Ready to begin low rate production.
Production and Deployment	9	Low rate production demonstrated. Capability in place to begin full rate production.
Operation and Support	10	Full rate production demonstrated and lean production practices in place.

Fig. Manufacturing Readiness Levels